

B.Tech. Degree I Semester/II Semester Supplementary Examination November 2019

CS/EC/IT AS 15-1102 B & CE/EE/ME/SE AS 15-1202 A ENGINEERING CHEMISTRY
(2015 Scheme)

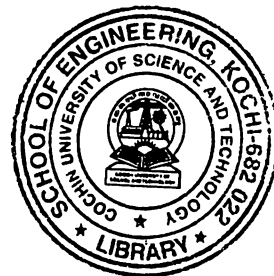
Time : 3 Hours

Maximum Marks : 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) Diamond is hard and bright. Explain why?
 (b) Give the Born-Haber cycle for determining the lattice energy of ionic crystals.
 (c) Illustrate and explain the fundamental modes of vibration of water molecule.
 (d) Explain how HCl molecule is microwave active.
 (e) Derive Gibbs-Helmholtz equation.
 (f) State and explain phase rule.
 (g) Explain the buffer action of a basic buffer.
 (h) Derive Nernst equation for single electrode potential.
 (i) Explain with examples the compounding of plastics.
 (j) Give the coordination mechanism of polymerization.



PART B

(4 × 10 = 40)

- II. (a) Write down the assumptions of free electron theory of metallic bonding. How the metallic luster, thermal and electrical conductivity of metals can be explained by this theory. (5)
 (b) Discuss the theory behind the microwave spectroscopy. Describe the method to determine the bond length of a diatomic molecule from microwave spectrum. (5)

OR

- III. (a) Discuss the formation of bands and Fermi level in metal atoms by band theory with a suitable example. (5)
 (b) Explain BET adsorption isotherm. How is it used to determine the surface area of a solid catalyst? (5)
- IV. (a) What are concentration cells? Derive Nernst equation for a concentration cell. (5)
 (b) Derive integrated rate law for a first order reaction and obtain an expression for the half-life period. (5)

OR

- V. (a) Explain the electrochemical process of rusting of Iron using a suitable figure. (5)
 (b) Describe the cathodic protection methods to prevent corrosion. (5)
- VI. (a) Explain (i) Nernst heat theorem (ii) Thermodynamics of biological reactions. (5)
 (b) Derive a relation between free energy change and equilibrium constant of a reaction. (5)

OR

- VII. (a) Derive Kirchoff's equations. (5)
 (b) Explain condensed phase rule and interpret the phase diagram of simple eutectic system. (5)
- VIII. (a) Discuss the free radical mechanism of polymerization. (5)
 (b) Explain the preparation of vulcanized rubber and its advantages over natural rubber. (5)

OR

- IX. (a) Give the preparation, properties and uses of (i) Nylon 66 (ii) Silicon polymers. (5)
 (b) Explain two mechanisms of lubrication. (5)